

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			reversible reaction accept either of following • reaction can go both ways • products (of the reaction) can react to form (the original) reactants	1			1		
	(b)			vanadium(V) oxide / vanadium oxide accept V_2O_5 neutral answer – VO	1			1		
	(c)			step 1 (sulfur trioxide) added to / dissolved in concentrated sulfuric acid (to form oleum) (1) accept $SO_3 + \text{conc } H_2SO_4 \rightarrow H_2S_2O_7$ (ignore incorrect formula for oleum) step 2 add product of first reaction to water / product is diluted (to form sulfuric acid) (1) accept $H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$ (ignore incorrect formula for oleum)	2			2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)			award (2) for all five points plotted correctly (± 1 square) award (1) for any three points plotted correctly award (1) for smooth curve of best fit		2	1	3	3	
	(e)			decreases		1		1		
	(f)			400 to 515-525 °C ecf possible from line based on incorrectly plotted points			1	1	1	
				Question 6 total	4	3	2	9	4	0